

2 The photograph shows *Daphnia*, a freshwater animal.



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Magnification $\times 20$

Daphnia is transparent and the heart can be seen beating.

A student investigated the effect of caffeine concentration on the heart rate of *Daphnia*.

The student selected five *Daphnia*: A, B, C, D and E. The student measured the heart rate, in beats per minute (bpm), of each *Daphnia* in water.

This was repeated using five different concentrations of caffeine solution: 0.1%, 0.5%, 1.0%, 2.0% and 5.0%.

The table shows the data from this investigation.

<i>Daphnia</i>	Concentration of caffeine (%)					
	0.0	0.1	0.5	1.0	2.0	5.0
A	176	257	259	268	274	282
B	178	256	260	270	282	274
C	184	262	264	270	278	285
D	172	248	251	260	270	278
E	182	264	258	272	272	286
mean heart rate/bpm	178	261	258	267	275	

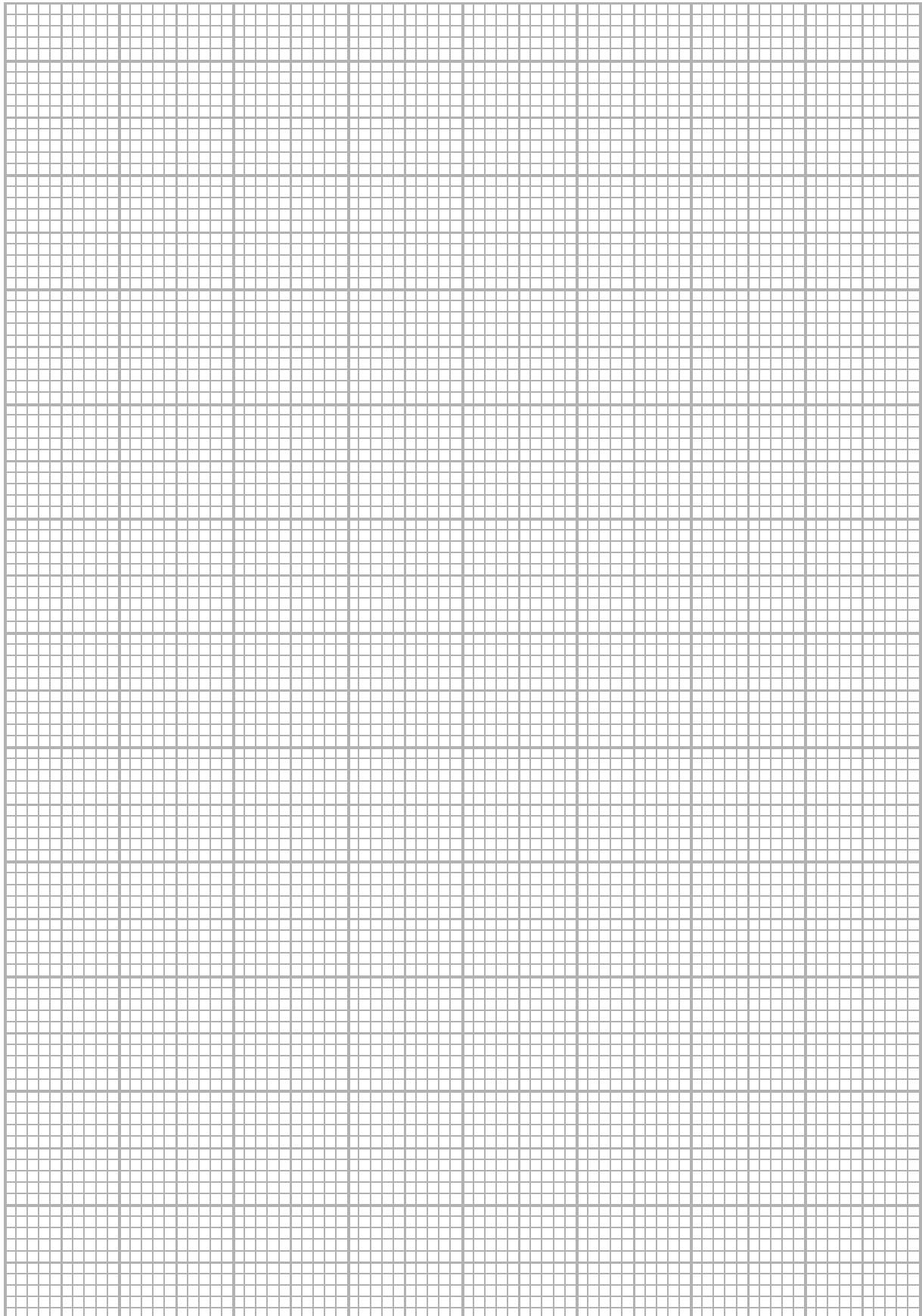
(a) Calculate the mean heart rate (bpm) for the 5.0% caffeine concentration.

(1)

Answer.....bpm

(b) Plot a suitable graph to show the effect of caffeine concentration on the mean heart rate of *Daphnia*.

(3)



(c) State a suitable null hypothesis for this investigation.

(2)

(d) Explain one ethical reason why the student chose to use *Daphnia* for this investigation.

(2)

- (e) The student used a statistical test to investigate the significance of the correlation between the mean heart rate and the caffeine concentration.

To calculate the correlation coefficient, the student produced the following table.

Caffeine concentration % (a)	Mean heart rate (b)	Rank a	Rank b	<i>d</i>
0.0	178	1	1	0
0.1	261	2	3	-1
0.5	258	3	2	1
1.0	267	4	4	0
2.0	275	5	5	0
5.0		6	6	0

- (i) Calculate the correlation coefficient, r_s , using the formula:

(3)

$$r_s = 1 - \frac{6\sum d^2}{n(n^2 - 1)}$$

Where:

Σ = the sum of

d = the difference between each pair of ranks

n = the size of the sample (number of pairs of values)

Answer.....

(ii) The table shows some critical values for this statistical test.

Number of pairs of values	Level of significance (p)		
	0.10	0.05	0.01
4	1.000	–	–
5	0.900	1.000	–
6	0.829	0.886	1.000
7	0.714	0.786	0.929
8	0.643	0.738	0.881
9	0.600	0.700	0.833
10	0.564	0.648	0.794

Explain how the student could use the graph and the statistical test to draw conclusions from this investigation.

(5)

(Total for Question 2 = 16 marks)